

Programming Using C++ Level 1

Switch Case Statement



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Definition

- ◆ A switch statement is a conditional statement used to choose one action from multiple possible cases.
- ◆ switch compares one expression with different case values.
- ◆ break stops the switch statement from continuing to the next case.
- ◆ default runs when none of the case values matches.

Explanation

- ◆ This program uses `grade / 10` to reduce the grade to a value between 0 and 10.
- ◆ The switch statement then checks this value.
 - 10 or 9 → A
 - 8 → B
 - 7 → C
 - 6 → D
 - 5 → E
 - Anything else → F
- ◆ Integer division is important here because `grade / 10` removes the decimal part.

Steps in main()

1. Get the grade

- ◆ Read the user's grade and store it in grade.

2. Divide the grade by 10

- ◆ Use grade / 10 to determine the range.

3. Check the cases

- ◆ Compare the result with the available case values.

4. Display the grade

- ◆ Print the corresponding letter for the matching case.

5. Stop the case

- ◆ Use break after each case.

6. Handle other values

- ◆ Use default when no case matches.

💡 Example

◆ If the user enters:

85

◆ The expression:

$\text{grade} / 10$

◆ becomes:

$85 / 10 = 8$

◆ The switch finds:

case 8:

◆ The program prints:

B

➡ Result

◆ Example results:

95 → A

85 → B

75 → C

65 → D

55 → E

45 → F

Note

- ◆ `grade / 10` uses integer division because `grade` is an `int`.
- ◆ `case 10` and `case 9` both produce `A` because grades from 90 to 100 become 9 or 10 after division by 10.
- ◆ `break` prevents execution from continuing into the next case.
- ◆ `default` acts as the fallback case.

⊘ Warning

- ◆ This program assumes that grades below 50 should produce F.
- ◆ It does not explicitly reject values greater than 100 or negative values.



Comparison

switch

- ✓ Useful when checking one expression against multiple fixed values.
- ✓ Makes multiple-choice conditions easier to read.
- ✓ Uses case and break to control execution.

if / else if

- ✓ Can handle complex conditions and ranges directly.
- ✓ More flexible for conditions such as grade ≥ 90 .
- ✓ Useful when conditions are not simple fixed values.

✦✦ Summary

- ✓ switch allows a program to choose between multiple cases.
- ✓ break stops execution after a matching case.
- ✓ Integer division can be used to convert grade ranges into values suitable for switch.

Thank you 😊